# Sri Lanka Institute of Information Technology




## Lab Submission
## <Lab sheet 03>

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**Discrete Mathematics | IT1160**

B.Sc. (Hons) in Information Technology

Part 01)

01.

In [18]:
```
def add_two_numbers(a,b):
    return(a+b)

print(add_two_numbers(5,7))
```

```
12
```

02.

--

In [17]:
```
def simple_interest(principal,rate,time):
    return(principal*rate*time)/100

print(simple_interest(1000,5,2))
```

```
100.0
```

03.

In [10]:
```
def assign_ip_address(device_ids):
    ip_map = {}
    base_ip = "192.168.1."
    last_octet = 2

    for device_id in device_ids:
        ip_map[device_id] = base_ip + str(last_octet)
        last_octet += 1

    return ip_map


devices = [501, 502, 503, 504]

ip_assignments = assign_ip_address(devices)

for device, ip in ip_assignments.items():
    print(f"Device ID {device} -> IP Address {ip}")
```

```
Device ID 501 -> IP Address 192.168.1.2
Device ID 502 -> IP Address 192.168.1.3
Device ID 503 -> IP Address 192.168.1.4
Device ID 504 -> IP Address 192.168.1.5
```

04.

In [12]:
```
def grade(mark):
    if 75 <= mark <= 100:
        return "A"
    elif 60 <= mark <= 74:
        return "B"
    elif 40 <= mark <= 59:
        return "C"
    else:
        return "F"

print(grade(89))
print(grade(38))
print(grade(67))
```

```
A
F
B
```

05.

In [12]:
```
import math

def f(x):
    if x == 0:
        raise ValueError("x cannot be 0")
    return 1 / x


def g(x):
    if x < 0:
        raise ValueError("x must be greater than or equal to 0")
    return x ** 0.5


def h(x):
    if x <= 0:
        raise ValueError("x must be greater than 0")
    return math.log(x)


def k(x):
    return x ** 2 + 3 * x + 2
```

In [16]:
```
def is_in_domain(func, x):
    try:
        func(x)
        return True
    except ValueError:
        return False

print(is_in_domain(f,0))
print(is_in_domain(g,2))
print(is_in_domain(h,-1))
print(is_in_domain(k,8))
```

```
False
True
False
True
```

Part 02)

01.

```
True
```

In [17]:
```
def area_of_rectangle(length, width):
    return length * width

print(area_of_rectangle(5, 3))
```

```
15
```

02.

```
15
```

In [19]:
```
def convert_to_fahrenheit(celsius):
    return (celsius * 9/5) + 32

print(convert_to_fahrenheit(0))
```

```
32.0
```

03.

In [20]:
```
def count_vowels(s):
    count = 0

    for vovel in s:
        if vovel in "aeiou":
            count += 1

    return count

print(count_vowels("hello world"))
```

```
3
```

04.

In [25]:
```
def encode(x):
    return 2*x + 3

def decode(x):
    return (x - 3)/2

pin = 1945

encoded = encode(pin)
decoded = decode(encoded)

print("Encoded:", encoded)
print("Decoded:", decoded)
print("Original PIN:", pin)
```

```
Encoded: 3893
Decoded: 1945.0
Original PIN: 1945
```

05.

In [29]:
```
functions = [p, q, r]

print(evaluate_functions(functions, 4))
```

```
[49, 0.5, 1.6094379124341003]
```

In [30]:
```
print(evaluate_functions(functions, -2))
```

```
[1, 'Domain Error', 'Domain Error']
```

In [27]:
```python
def p(x):
    return x**3 - 4*x + 1

def q(x):
    if x <= 0:
          raise ValueError("x cannot be equal or lesss than 0")
    return 1/(x**0.5)

import math

def r(x):
    if x <= -1:
        raise ValueError("x must be greater than -1")
    return math.log(x + 1)
```

In [28]:
```python
def evaluate_functions(func_list, x):
    results = []

    for func in func_list:
        try:
            results.append(func(x))
        except ValueError:
            results.append("Domain Error")

    return results
```